Title:

Products with residual RFI degradation acquired in February 2024

Description:

Since the deployment of the SAR Processor version IPF 3.51, a process of pre-screening and mitigation of RFI (Radio Frequency Interference) is in place. The pre-screening is based on available noise-only pulses, while the mitigation involves the detection and correction of RFI artefacts in the SAR data. The process does not allow for a complete mitigation of all RFI impacting the quality of the Sentinel-1 products, either due to mis detection at pre-screening or due to unsuccessful data-based detection/correction.

The residual RFIs are monitored via quick-look inspection and the annotations of the impacted data are evaluated to categorize the residuals in:

- Products with RFI mitigation "on" and observed residual RFI: in this case, RFI was pre-screened in at least one sub-swath, the RFI module was activated, but the data-based detection/correction wasn't 100% effective.
- Products with RFI mitigation "off" but with observed RFI: in this case, the pre-screening did not report any RFI, but RFI was observed on the quick-look.

The purpose of this quality disclaimer is to provide a list of products impacted by such residual degradation in the reporting period (as indicated in the title). Moreover, this disclaimer provides the number of products for which the processor detected and mitigated RFI and the categorization according to the annotation, also limited to the reporting period. The list of products with residual RFI is provided at the end of this report. The degradation is mainly on the radiometry for the parts of the images affected by the residual RFI. The table below provides occurrences statistics for the reporting period:

	EW_GRDM & IW_SLC	EW_GRDM	IW_SLC
Products acquired	17907 (100%)	2092 (100%)	15815 (100%)
Products with RFI mitigation "on"	3921 (21.9%)	28 (1.3%)	3893 (24.6%)
Products with observed residual RFI	192 (1.1%)	2 (0.1%)	187 (1.2%)
Products with RFI mitigation "off" but with observed RFI	139 (0.8%)	2 (0.1%)	134 (0.8%)
Products with RFI mitigation "on" and observed residual RFI	53 (0.3%)	0 (0.0%)	53 (0.3%)

Degradation types:

- ☒ DEGRADED_PRODUCT_RADIOMETRY
- ☐ DEGRADED_RADIOMETRIC_CALIBRATION
- ☐ DEGRADED_ORBIT_CONTROL
- ☐ COMPLETE_PRODUCT_DEGRADATION
- ☒ DEGRADED_PHASE
- ☐ DEGRADED_PRODUCT_GEOLOCATION
- ☐ DEGRADED_PLATFORM_POINTING
- ☐ DEGRADED_PERFORMANCE_INSTRUMENT_ANOMALY
- ☐ SLICE_PRODUCT_NON_CONCATENABLE
- ☐ OTHER

Degradation percentage[1]:

1%

Only a subset of the products acquired in the period are affected.

Only parts of the affected products are impacted by the degradation.

Impacted products:

Platform: ☒ S-1A ☐ S-1B ☐ S-1C

Acquisition mode: ☒ EW ☒ IW ☒ SM ☐ WV ☐ RF

Product type: ☐ RAW ☒ SLC ☒ GRDM ☒ GRDH ☒ GRDF ☒ OCN

Polarization: ☒ SH (Single pol. H) ☒ SV (Single pol. V) ☒ DH (Double pol. H) ☒ DV (Double pol. V)

1 Percentage of degradation of the data in the product (100% means that the product should be masked in the product catalogue)

Processing facility: N/A
IPF version: N/A
Instrument Configuration ID (RDB): N/A
ADF files:

AUX_INS	N/A
AUX_CAL	N/A
AUX_PP1	N/A
AUX_PP2	N/A
AUX_SCS	N/A

Period of the issue:

	Start	Stop
Acquisition date	2024-02-01T00:00:00	2024-03-01T00:00:00
Generation date	2024-02-01T00:00:00	2024-03-02T00:00:00
Orbit	52358	52778
Datatake (hex)	0654C4	066313

Cause:

Sentinel-1 is transmitting electromagnetic pulses in C-Band. This frequency band is shared with other emitters that can interfere with the signal received by the Sentinel-1 SAR, causing Radio Frequency Interference (RFI).
Since IPF 3.51, the SAR processing implements a pre-screening and a mitigation of the RFI. However, this pre-screening/mitigation does not allow removing all the interferences in the SAR product.
Residual RFI can then be observed on SAR images.

Status:

The number of Sentinel-1 products for which the IPF detects and mitigate an RFI is monitored using the product annotation.
The products for which residual RFI are observable are monitored via quick-look inspection.
Usually, the RFI only impacts a portion of the surface covered by the product, and then the spatial coverage of the degradation is limited. Examples of residual RFI for the period is presented below.

A list of products impacted by residual RFI is provided below as a pattern of product names. Note that:

- This list may not be exhaustive as small RFI degradation may not be visible on the quick-looks.
- The patterns provide mainly information on acquisition mode and start/stop of acquisition time.
- When a product type (GRD for instance) is impacted then the other types are impacted as well (SLC, OCN)
- This list is provided considering the nominal processing flow and the generation date (table above) is set accordingly. No systematic reprocessing of those products is foreseen for now and even so, it cannot be guaranteed that a reprocessing could perform and efficient RFI mitigation for those specific products.

References:

- **MPC ref:** MPCS-3658

Example of residual RFI:

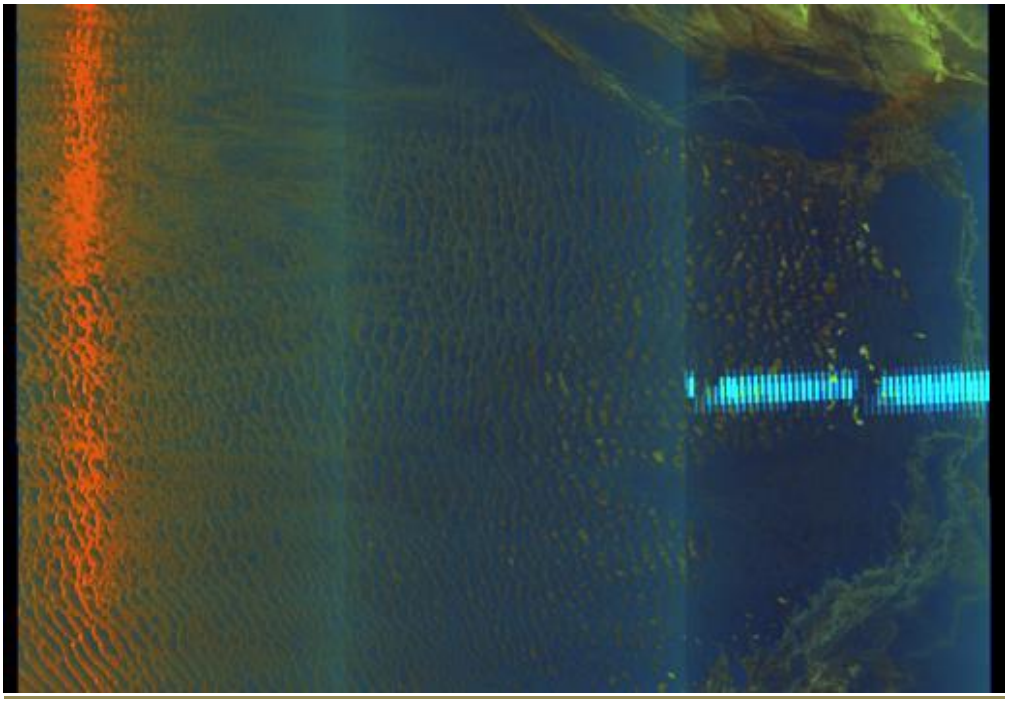


Figure 1: S1A_IW_GRDH_1ADV_20240211T120918_20240211T120943_052511_0659E9_060E



Figure 2: S1A_IW_GRDH_1ADV_20240215T052502_20240215T052527_052565_065BBA_E141

List of products with RFI mitigation “off” but observed RFI (product name templates)

S1A_IW_*_*DV_20240201T053609_20240201T053638_052361_0654DA_*
S1A_IW_*_*DV_20240201T132826_20240201T132851_052366_065502_*
S1A_IW_*_*DV_20240201T151224_20240201T151249_052367_06550E_*
S1A_IW_*_*DV_20240201T183246_20240201T183311_052369_06551F_*
S1A_IW_*_*DV_20240201T222322_20240201T222351_052371_065537_*
S1A_IW_*_*DV_20240202T044916_20240202T044941_052375_065552_*
S1A_IW_*_*DV_20240202T062738_20240202T062803_052376_06555E_*
S1A_IW_*_*DV_20240202T075809_20240202T075838_052377_065568_*
S1A_IW_*_*DV_20240202T153912_20240202T153941_052381_065595_*
S1A_IW_*_*DV_20240202T154031_20240202T154056_052381_065595_*
S1A_IW_*_*DV_20240202T154056_20240202T154121_052381_065595_*
S1A_IW_*_*DV_20240202T155249_20240202T155314_052382_065596_*
S1A_IW_*_*DV_20240202T155822_20240202T155847_052382_065597_*
S1A_IW_*_*DV_20240202T173450_20240202T173515_052383_06559F_*
S1A_IW_*_*DV_20240203T113722_20240203T113747_052394_0655F5_*
S1A_IW_*_*DV_20240203T131648_20240203T131713_052395_0655FC_*
S1A_IW_*_*DV_20240204T025126_20240204T025151_052403_065647_*
S1A_IW_*_*DV_20240204T025151_20240204T025216_052403_065647_*
S1A_IW_*_*DV_20240204T042654_20240204T042719_052404_065650_*
S1A_IW_*_*DV_20240204T111201_20240204T111226_052408_06567A_*
S1A_IW_*_*DV_20240204T153724_20240204T153749_052411_06568D_*
S1A_IW_*_*DV_20240204T171248_20240204T171313_052412_06569A_*
S1A_IW_*_*DV_20240204T171838_20240204T171903_052412_06569A_*
S1A_IW_*_*DV_20240204T185743_20240204T185812_052413_0656A4_*
S1A_IW_*_*DV_20240204T185812_20240204T185837_052413_0656A4_*
S1A_EW_*_*DH_20240202T191452_20240202T191551_052384_0655AC_*
S1A_IW_*_*DV_20240205T015404_20240205T015429_052417_0656C9_*
S1A_IW_*_*DV_20240205T175640_20240205T175705_052427_065720_*
S1A_IW_*_*DV_20240205T214740_20240205T214805_052429_06572F_*
S1A_IW_*_*DV_20240205T215035_20240205T215100_052429_06572F_*
S1A_IW_*_*DV_20240206T023636_20240206T023701_052432_065742_*
S1A_IW_*_*DV_20240206T184333_20240206T184407_052442_065794_*
S1A_IW_*_*DV_20240207T174322_20240207T174347_052456_06580C_*
S1A_IW_*_*DV_20240207T194112_20240207T194137_052457_06581D_*
S1A_IW_*_*DV_20240208T022951_20240208T023016_052461_06582F_*
S1A_IW_*_*DV_20240208T053225_20240208T053250_052463_065842_*
S1A_IW_*_*DV_20240208T135433_20240208T135458_052468_065876_*
S1A_IW_*_*DV_20240208T182420_20240208T182445_052471_06588E_*
S1A_S2_*_*DV_20240208T014918_20240208T014937_052461_06582B_*
S1A_IW_*_*DV_20240207T160051_20240207T160116_052455_065802_*
S1A_IW_*_*DV_20240208T221440_20240208T221509_052473_0658A4_*
S1A_IW_*_*DV_20240208T221509_20240208T221534_052473_0658A4_*
S1A_IW_*_*DV_20240209T140339_20240209T140404_052483_0658F6_*
S1A_IW_*_*DV_20240209T140519_20240209T140544_052483_0658F6_*
S1A_IW_*_*DV_20240209T154445_20240209T154510_052484_065900_*
S1A_IW_*_*DV_20240210T020327_20240210T020352_052490_06592E_*
S1A_IW_*_*DV_20240210T020622_20240210T020647_052490_06592E_*
S1A_IW_*_*DV_20240210T051513_20240210T051538_052492_06593E_*
S1A_IW_*_*DV_20240210T051628_20240210T051653_052492_06593E_*
S1A_IW_*_*DV_20240210T051653_20240210T051718_052492_06593E_*
S1A_IW_*_*DV_20240210T094822_20240210T094851_052495_06595E_*
S1A_IW_*_*DV_20240210T144657_20240210T144722_052498_065978_*
S1A_IW_*_*DV_20240210T180845_20240210T180910_052500_065989_*
S1A_IW_*_*DV_20240211T055821_20240211T055846_052507_0659C1_*
S1A_IW_*_*DV_20240211T120918_20240211T120943_052511_0659E9_*
S1A_IW_*_*DV_20240211T152728_20240211T152753_052513_065A01_*
S1A_IW_*_*DV_20240211T171247_20240211T171312_052514_065A09_*
S1A_IW_*_*DV_20240211T184936_20240211T185005_052515_065A14_*
S1A_IW_*_*DV_20240211T185030_20240211T185055_052515_065A14_*
S1A_IW_*_*DV_20240212T045854_20240212T045919_052521_065A3C_*

S1A_IW_*_*DV_20240212T093202_20240212T093227_052524_065A60_*
S1A_IW_*_*DV_20240212T142923_20240212T142948_052527_065A7B_*
S1A_IW_*_*DV_20240212T175305_20240212T175330_052529_065A8B_*
S1A_IW_*_*DV_20240213T054022_20240213T054047_052536_065ABD_*
S1A_IW_*_*DV_20240213T183310_20240213T183335_052544_065B02_*
S1A_EW_*_*DH_20240213T152635_20240213T152739_052542_065AF3_*
S1A_IW_*_*DV_20240214T155911_20240214T155936_052557_065B73_*
S1A_IW_*_*DV_20240214T155846_20240214T155911_052557_065B73_*
S1A_IW_*_*DV_20240214T172810_20240214T172835_052558_065B7B_*
S1A_IW_*_*DV_20240214T173515_20240214T173540_052558_065B7B_*
S1A_IW_*_*DV_20240214T235641_20240214T235710_052562_065BA1_*
S1A_IW_*_*DV_20240215T021142_20240215T021207_052563_065BA9_*
S1A_IW_*_*DV_20240215T034923_20240215T034948_052564_065BB2_*
S1A_IW_*_*DV_20240215T052502_20240215T052527_052565_065BBA_*
S1A_IW_*_*DV_20240215T134546_20240215T134611_052570_065BE6_*
S1A_IW_*_*DV_20240215T181636_20240215T181701_052573_065C00_*
S1A_IW_*_*DV_20240216T025151_20240216T025216_052578_065C30_*
S1A_IW_*_*DV_20240216T025216_20240216T025241_052578_065C30_*
S1A_IW_*_*DV_20240216T025931_20240216T025956_052578_065C31_*
S1A_IW_*_*DV_20240216T135710_20240216T135735_052585_065C71_*
S1A_IW_*_*DV_20240216T171813_20240216T171838_052587_065C86_*
S1A_IW_*_*DV_20240216T185743_20240216T185812_052588_065C90_*
S1A_IW_*_*DV_20240216T185837_20240216T185902_052588_065C90_*
S1A_IW_*_*DV_20240217T015334_20240217T015403_052592_065CB3_*
S1A_IW_*_*DV_20240217T160433_20240217T160458_052600_065CFD_*
S1A_IW_*_*DV_20240217T162146_20240217T162211_052601_065CFE_*
S1A_IW_*_*DV_20240217T162236_20240217T162301_052601_065CFE_*
S1A_IW_*_*DV_20240217T180000_20240217T180025_052602_065D04_*
S1A_IW_*_*DV_20240218T024454_20240218T024519_052607_065D24_*
S1A_IW_*_*DV_20240218T024634_20240218T024659_052607_065D24_*
S1A_IW_*_*DV_20240218T024724_20240218T024752_052607_065D24_*
S1A_IW_*_*DV_20240218T151137_20240218T151202_052615_065D67_*
S1A_IW_*_*DV_20240218T151621_20240218T151646_052615_065D68_*
S1A_IW_*_*DV_20240218T152121_20240218T152146_052615_065D68_*
S1A_IW_*_*DV_20240218T170405_20240218T170430_052616_065D72_*
S1A_IW_*_*DV_20240218T170430_20240218T170455_052616_065D72_*
S1A_S4_*_*DH_20240216T234919_20240216T234943_052591_065CAA_*
S1A_IW_*_*DV_20240218T204420_20240218T204433_052618_065D86_*
S1A_IW_*_*DV_20240219T142226_20240219T142251_052629_065DE3_*
S1A_IW_*_*DV_20240219T154816_20240219T154841_052629_065DEC_*
S1A_IW_*_*DV_20240219T173757_20240219T173822_052631_065DF8_*
S1A_IW_*_*DV_20240219T174232_20240219T174257_052631_065DF8_*
S1A_IW_*_*DV_20240219T174322_20240219T174347_052631_065DF8_*
S1A_IW_*_*DV_20240219T213054_20240219T213123_052633_065E11_*
S1A_IW_*_*DV_20240220T035750_20240220T035815_052637_065E28_*
S1A_IW_*_*DV_20240220T040045_20240220T040110_052637_065E28_*
S1A_IW_*_*DV_20240220T145419_20240220T145448_052644_065E6B_*
S1A_S2_*_*DV_20240220T014918_20240220T014937_052636_065E1E_*
S1A_IW_*_*DV_20240220T221649_20240220T221714_052648_065E96_*
S1A_IW_*_*DV_20240221T140015_20240221T140044_052658_065EE1_*
S1A_IW_*_*DV_20240221T153311_20240221T153345_052659_065EEA_*
S1A_IW_*_*DV_20240221T155046_20240221T155111_052659_065EEC_*
S1A_IW_*_*DV_20240221T172734_20240221T172759_052660_065EF6_*
S1A_IW_*_*DV_20240221T185917_20240221T185942_052661_065EFF_*
S1A_IW_*_*DV_20240222T051448_20240222T051513_052667_065F36_*
S1A_IW_*_*DV_20240222T051603_20240222T051628_052667_065F36_*
S1A_IW_*_*DV_20240222T052218_20240222T052243_052667_065F36_*
S1A_IW_*_*DV_20240222T144657_20240222T144722_052673_065F6E_*
S1A_IW_*_*DV_20240222T163048_20240222T163113_052674_065F77_*
S1A_IW_*_*DV_20240222T193955_20240222T194020_052676_065F8A_*
S1A_IW_*_*DV_20240222T194020_20240222T194045_052676_065F8A_*

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S1A_IW_*_*DV_20240223T042445_20240223T042510_052681_065FAE_*
S1A_IW_*_*DV_20240223T060411_20240223T060435_052682_065FB5_*
S1A_IW_*_*DV_20240223T102625_20240223T102650_052685_065FD0_*
S1A_IW_*_*DV_20240223T184936_20240223T185005_052690_066007_*
S1A_IW_*_*DV_20240223T205216_20240223T205242_052691_066011_*
S1A_IW_*_*DV_20240224T142304_20240224T142333_052702_06606D_*
S1A_IW_*_*DV_20240224T175125_20240224T175150_052704_06607D_*
S1A_IW_*_*DV_20240226T153912_20240226T153941_052731_066166_*
S1A_IW_*_*DV_20240226T154031_20240226T154056_052731_066166_*
S1A_IW_*_*DV_20240227T112837_20240227T112902_052744_0661D6_*
S1A_IW_*_*DV_20240227T131557_20240227T131622_052745_0661DB_*
S1A_IW_*_*DV_20240227T234055_20240227T234120_052751_06621A_*
S1A_IW_*_*DV_20240228T030226_20240228T030251_052753_066225_*
S1A_IW_*_*DV_20240228T185812_20240228T185837_052763_06627C_*
S1A_IW_*_*DV_20240228T185837_20240228T185902_052763_06627C_*
S1A_IW_*_*DV_20240228T185952_20240228T190020_052763_06627C_*
S1A_IW_*_*DV_20240229T050636_20240229T050701_052769_0662B7_*
S1A_IW_*_*DV_20240229T143848_20240229T143913_052775_0662F2_*
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List of products with RFI mitigation “on” but observed RFI (product name templates)

S1A_IW_*_*DV_20240201T235222_20240201T235247_052372_06553B_*
S1A_IW_*_*DV_20240202T031450_20240202T031519_052374_06554C_*
S1A_IW_*_*DV_20240202T075838_20240202T075903_052377_065568_*
S1A_IW_*_*DV_20240202T075903_20240202T075918_052377_065568_*
S1A_IW_*_*DV_20240202T140944_20240202T141009_052381_06558C_*
S1A_IW_*_*DV_20240202T153941_20240202T154006_052381_065595_*
S1A_IW_*_*DV_20240202T154814_20240202T154839_052382_065596_*
S1A_IW_*_*DV_20240202T211729_20240202T211754_052385_0655B3_*
S1A_IW_*_*DV_20240203T021027_20240203T021052_052388_0655C4_*
S1A_IW_*_*DV_20240203T021257_20240203T021322_052388_0655C4_*
S1A_IW_*_*DV_20240203T021944_20240203T022009_052388_0655C5_*
S1A_IW_*_*DV_20240205T050702_20240205T050727_052419_0656D8_*
S1A_IW_*_*DV_20240205T143553_20240205T143618_052425_065711_*
S1A_IW_*_*DV_20240205T143733_20240205T143758_052425_065711_*
S1A_IW_*_*DV_20240206T023906_20240206T023931_052432_065742_*
S1A_IW_*_*DV_20240206T120241_20240206T120306_052438_065771_*
S1A_IW_*_*DV_20240206T151826_20240206T151851_052440_065782_*
S1A_IW_*_*DV_20240207T155522_20240207T155551_052455_065802_*
S1A_IW_*_*DV_20240208T015016_20240208T015045_052461_06582C_*
S1A_IW_*_*DV_20240208T040045_20240208T040110_052462_065838_*
S1A_IW_*_*DV_20240208T100223_20240208T100248_052466_065863_*
S1A_IW_*_*DV_20240208T234904_20240208T234930_052474_0658AB_*
S1A_IW_*_*DV_20240210T143902_20240210T143927_052498_065978_*
S1A_IW_*_*DV_20240210T143927_20240210T143952_052498_065978_*
S1A_IW_*_*DV_20240211T011114_20240211T011139_052504_0659AC_*
S1A_IW_*_*DV_20240211T204844_20240211T204909_052516_065A1D_*
S1A_IW_*_*DV_20240213T023219_20240213T023244_052534_065AAE_*
S1A_IW_*_*DV_20240213T164940_20240213T165005_052543_065AFA_*
S1A_IW_*_*DV_20240213T165415_20240213T165440_052543_065AFA_*
S1A_IW_*_*DV_20240215T163745_20240215T163810_052572_065BF6_*
S1A_IW_*_*DV_20240216T043309_20240216T043334_052579_065C39_*
S1A_IW_*_*DV_20240216T135206_20240216T135235_052585_065C71_*
S1A_IW_*_*DV_20240216T152810_20240216T152844_052586_065C77_*
S1A_IW_*_*DV_20240216T152959_20240216T153028_052586_065C78_*
S1A_IW_*_*DV_20240217T143114_20240217T143143_052600_065CF5_*
S1A_IW_*_*DV_20240217T143758_20240217T143823_052600_065CF5_*
S1A_IW_*_*DV_20240217T231817_20240217T231846_052605_065D18_*
S1A_IW_*_*DV_20240218T151043_20240218T151112_052615_065D67_*
S1A_IW_*_*DV_20240219T154956_20240219T155031_052630_065DEC_*
S1A_IW_*_*DV_20240222T051628_20240222T051653_052667_065F36_*
S1A_IW_*_*DV_20240222T143902_20240222T143927_052673_065F6E_*
S1A_IW_*_*DV_20240223T152843_20240223T152908_052688_065FF4_*
S1A_IW_*_*DV_20240224T110454_20240224T110519_052700_06605C_*
S1A_IW_*_*DV_20240224T161427_20240224T161452_052703_066073_*
S1A_IW_*_*DV_20240226T030635_20240226T030700_052724_06611D_*
S1A_IW_*_*DV_20240226T154056_20240226T154121_052731_066166_*
S1A_IW_*_*DV_20240226T154814_20240226T154839_052732_066167_*
S1A_IW_*_*DV_20240228T153224_20240228T153249_052761_066265_*
S1A_IW_*_*DV_20240229T033637_20240229T033702_052768_0662AE_*
S1A_IW_*_*DV_20240229T143114_20240229T143143_052775_0662F2_*
S1A_IW_*_*DV_20240229T143708_20240229T143733_052775_0662F2_*
S1A_IW_*_*DV_20240229T224544_20240229T224609_052780_066317_*
S1A_IW_*_*DV_20240229T232526_20240229T232551_052780_06631B_*